

Spenny Piggy Infrastructure & Scaling Plan (Final)

Objective:

Build a scalable, high-performance platform that starts lean at low cost and automatically scales with demand, targeting a maximum baseline infrastructure cost of ~£3,500/month at full load.

Target Scale (Design For):

- 1,000 active creators
- 100,000 total users
- 20,000 daily active users (DAU)
- 1,000–2,000 concurrent users (design for 2,000+)

Performance Requirements:

- Page load time: < 1.5 seconds
- API response time: < 200ms (p95)
- Payment actions: < 1 second perceived latency

Traffic Assumptions:

- 20–50 requests per user per minute during active sessions
- High load from payments, messaging, and notifications

Core Requirement: Elastic (Auto-Scaling) Architecture

- Infrastructure must automatically scale UP during high demand
- Infrastructure must scale DOWN during low usage to minimise cost
- Backend must be stateless to support horizontal scaling

Cost Strategy:

- DO NOT build a fixed £3,500/month infrastructure from day one
- Current acceptable cost: ~£100–£300/month during build/early usage
- Infrastructure should scale gradually with usage:
 - Early stage: £100–£300/month
 - Initial growth: £300–£800/month
 - Scaling phase: £800–£2,000/month
 - Target scale: up to ~£3,500/month

Infrastructure Requirements:

- Auto-scaling compute (e.g. containers or serverless)
- Managed scalable database
- CDN for all static assets

- Caching layer (e.g. Redis) for performance
- Queue system for background jobs (payments, notifications, Stripe webhooks)

Critical Systems:

- Stripe webhook handling MUST be reliable and queued
- Payment processing must be resilient to spikes
- No blocking operations on user-facing requests

Scalability Goal:

System should handle 10x growth without major refactor:

- 10,000 creators
- 1,000,000 users
- 10,000+ concurrent users

Summary:

Build lean, scale automatically, and ensure reliability under payment-heavy workloads. The system must prioritise performance, cost efficiency, and the ability to grow without re-architecture.